

VAEREN STUDIOS

V.02 ARMORED ESSENTIAL HOODIE

MINIMAL EDITION | PRODUCTION TECH PACK | REV 01 | 2026-09-06

VAEREN STUDIOS
DESIGN FOR A DIFFERENT TOMORROW
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V.02
ARMORED ESSENTIAL HOODIE
MINIMAL EDITION

FORM
FUNCTION
IDENTITY



FRONT
0°

FRONT ANGLE
45°

SIDE
90°

BACK
180°

BACK ANGLE
225°

OPPOSITE SIDE
270°

01. HOOD & PLACKET
HALF ZIP / CLEAN FINISH

02. PANEL DETAIL
MINIMAL RIVET HARDWARE

03. RAW EDGE
DISTRESSED FINISH

04. CUFF
RIB KNIT

05. FABRIC
GARMENT WASHED FRENCH TERRY

06. BRANDING
SUBTLE BACK EMBROIDERY

TECHNICAL FLATS (NOT TO SCALE)

FRONT
BACK
LEFT SIDE
RIGHT SIDE

MEASUREMENTS (SIZE M)


POINT OF MEASURE (CM)	M
A. CHEST WIDTH	64
B. BODY LENGTH	58
C. SHOULDER WIDTH	62
D. SLEEVE LENGTH	56
E. HEM WIDTH	52

FIT

BOX FIT
CROPPED LENGTH
DROP SHOULDER

FABRIC

100% COTTON
400-450 GSM
FRENCH TERRY
GARMENT WASHED

COLORWAYS

WASHED BLACK

WASHED GREY

MASHED BROWN

WASHED OLIVE

HARDWARE

YKK ZIPPER

CUSTOM RIVET

LABELING

MAIN LABEL

SIZE TAB

CARE LABEL

PACKAGING

PACKAGING

VAEREN STUDIOS

A NEW FORM OF EXPRESSION

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Factory-use note: This pack translates the supplied design board into a production-facing BOM, POM/grading, construction, branding, wash, QC and approval package. The board's M base measurements are treated as confirmed references; secondary dimensions are explicitly marked provisional and must be locked after proto fitting.

01 - DESIGN REFERENCE & TECHNICAL FLATS



DESIGN INTENT	NON-NEGOTIABLES	CRITICAL PROTO CHECKS
Box fit, cropped body, drop shoulder, structured hood, engineered arm overlays, washed premium surface.	Symmetry; panel geometry; zipper alignment; cuff/hem recovery; controlled wash; clean interior; burr-free hardware.	Hood volume; zipper torque/wave; shoulder width; arm-panel mobility; rib recovery; wash shrinkage; shade repeatability.

02 - BILL OF MATERIALS

Primary fabric: 100% cotton French terry, 400-450 GSM, garment washed. Hardware and trim are locked only after handloom / wash / pull approval.

Item	Component	Material / specification	Construction / finish	Color	Supplier / source	Placement	Qty per garment	Approval
01	Main shell	100% cotton French terry; 400-450 GSM; compact; target width 180-190 cm	Garment washed; controlled vintage effect; no plasticky handfeel	Washed Black primary	TBD	Body, hood, sleeves, arm overlays	TBD by marker	LAB DIP + WASH SWATCH
02	Rib	2x2 cotton rib; recommend 95/5 cotton-elastane; target 480-520 GSM	Tubular/knit; firm recovery; pre-shrunk	Match shell	TBD	Cuffs + hem	TBD by pattern	HAND-FEEL + RECOVERY
03	Hood facing / internal clean finish	Self fabric or 180-220 GSM cotton jersey, final after proto	Internal seam clean finish; avoid bulk at zipper	Black	TBD	Hood opening / zipper area	TBD	PROTO APPROVAL
04	Zipper	YKK #5 reverse-coil or equivalent; closed-end; black tape	Auto-lock slider; matte finish; custom cut length	Black	YKK / approved supplier	Center-front half zip	1	HARDWARE APPROVAL
05	Zipper pull	YKK standard pull or approved custom pull	Matte black/gunmetal; no sharp edges	Gunmetal / Black	YKK / TBD	Center-front zipper	1	HARDWARE APPROVAL
06	Custom rivet	Metal cap rivet, approx. 8 mm face diameter; post length to shell stack	Matte gunmetal / antique silver; burr-free	Gunmetal	TBD	Arm-panel stress points per approved sample	TBD after pattern	HARDWARE + PULL TEST
07	Sewing thread	Poly-core spun or equivalent; Tex 40-60 general / Tex 60 at stress points	High abrasion resistance; colorfast; match shell	Black	TBD	All seams	As required	SHADE MATCH
08	Main label	Damask/woven label; VAEREN STUDIOS + logo mark	Black ground; light grey/white lettering	Black / Grey	TBD label house	Inside back neck	1	ARTWORK APPROVAL
09	Size tab	Woven or satin printed size tab	Single letter; low-profile	Black / White	TBD	Below main label	1	ARTWORK APPROVAL
10	Care label	Satin printed care label; composition + country + care instructions	Use final legal content for destination market	White/Black or Black/White	TBD	Left side seam / approved location	1	LEGAL COPY APPROVAL
11	Brand polybag	Matte frosted PE/LDPE polybag with VAEREN STUDIOS mark	Self-adhesive flap; appropriate warning text per market	Frosted / Black print	TBD	Individual garment pack	1	PACKAGING APPROVAL
12	Interlining / reinforcement	Lightweight fusible only where required for zipper or rivet support	Soft hand; prevent show-through	Black	TBD	Front zipper / rivet backing only if needed	TBD	PROTO APPROVAL

Material-development caution: Rib composition, exact zipper gauge, rivet post length and hood internal facing are not specified in the supplied board. This pack provides workable starting recommendations, but those items must be approved on the first proto before bulk.

03 - MEASUREMENT SPECIFICATION & GRADING

POM	Measurement method	Tolerance (cm)	XS	S	M	L	XL	Grade rule / note
A. Chest width	1 cm below armhole; straight across body at relaxed state	±1.0	60	62	64	66	68	2 cm per size; total chest circumference approx. 4x width
B. Body length	HPS at neck to bottom edge of rib	±1.0	56	57	58	59	60	1 cm per size; preserve cropped proportion
C. Shoulder width	Shoulder tip to shoulder tip across back	±1.0	60	61	62	63	64	1 cm per size; drop-shoulder geometry
D. Sleeve length	Shoulder seam to sleeve/cuff join at natural sleeve line	±1.0	54	55	56	57	58	1 cm per size
E. Hem width	Straight across bottom hem, relaxed	±1.0	48	50	52	54	56	2 cm per size; total hem circumference approx. 4x width
F. Hood opening	Measure along finished face opening	±1.0	29	30	31	32	33	PROVISIONAL - set from approved hood pattern
G. Hood height	Center back neck to hood crown	±1.0	37	38	39	40	41	PROVISIONAL - set from approved hood pattern
H. Cuff height	Finished rib height at side seam	±0.5	7.0	7.0	7.5	7.5	8.0	PROVISIONAL
I. Hem rib height	Finished rib height center front/back	±0.5	7.0	7.0	7.5	7.5	8.0	PROVISIONAL
J. Zipper length	Top stop to bottom stop; center front	±0.5	27	28	29	30	31	PROVISIONAL - final after zipper placement on proto
K. Arm overlay length	Top of overlay to bottom point	±0.8	31	32	33	34	35	PROVISIONAL; verify proportion and mobility
L. Front overlay width	Maximum width of external armor panel	±0.5	13	13.5	14	14.5	15	PROVISIONAL; mirror left/right

Base-size evidence: A-E are taken directly from the supplied design board: M chest width 64 cm, body length 58 cm, shoulder width 62 cm, sleeve length 56 cm, hem width 52 cm. The board does not define the grading increments. The XS-XL grade rules here are proposed to preserve the reference proportions and require fit approval.

Measurement condition: garment relaxed, finished and after final wash/finish. Unless otherwise specified, measure flat with no stretch.

04 - CONSTRUCTION & SEWING SPEC

Operation	Construction requirement	Seam allowance / stitch	Topstitch	Reinforcement	QC / acceptance
01 - Shoulder / armhole	Follow approved flat exactly; no change to drop-shoulder seam angle after PP.	6-10 mm SA; 301 lockstitch; 4-5 SPI equivalent	Single needle 3-4 mm where visible	Backtack at starts/ends	Left/right symmetry ± 2 mm
02 - Body side seams	Closed seam; clean finish. No twisting after wash.	10 mm SA; 4-thread overlock + clean lockstitch where required	Optional 3 mm single needle on visible panel seam	Backtack / secure overlock	Seam pucker not acceptable
03 - Hood	2-piece hood with center-back seam; maintain crown volume shown in reference.	6-8 mm SA; 4-thread overlock	3 mm edge stitch around opening if used	Bartack/lock at zipper end area	No seam roping; center seam straight
04 - Half zipper / front placket	Center zipper must be straight and flat; zipper tape fully concealed or cleanly framed per reference.	6-8 mm SA; zipper foot; 301 stitch	2.5-3 mm edge topstitch along placket	Bar tack / box reinforcement at lower stop if needed	No wave, grin, twist or exposed tape beyond spec
05 - Armored overlay panels	Cut as mirrored left/right pairs. Maintain pointed lower silhouette and seam geometry.	6-8 mm SA; 301/overlock as pattern dictates	Double topstitch at external panel perimeter; 3-4 mm	Rivets at approved points; backing patch only if needed	Panel placement within ± 2 mm; rivet centered
06 - Raw-edge panel detail	Where a raw/distressed edge is intended, edge must be controlled, not frayed through stitch line.	Follow approved sample; stabilize seam line first	Visible distress limited to outer edge only	Do not remove structural seam allowance	No loose threads > 5 mm; no panel delamination
07 - Cuffs	Attach 2x2 rib evenly; rib slightly negative to opening for recovery.	10 mm join; overlock	Optional twin-needle cover at opening if approved	No additional hard reinforcement	Recovery after stretch: no permanent bagging
08 - Hem rib	Hem band joined with even tension; side seams must align.	10 mm join; overlock	Optional 3 mm cover/twin needle	None	Hem must lie flat and parallel
09 - Rivet installation	Pre-punch exact hole size; use die matched to hardware; protect fabric from distortion.	Factory set-up / press	N/A	Backing patch only if pull test fails	No sharp burr; no fabric tear; centered to panel
10 - Garment wash	Wash after full construction; recipe locked after pilot. Aim for dry hand, controlled fade, low contrast.	Use approved wash recipe	Do not over-soften	No excessive abrasion at rivets / zipper	Record pre/post measurements and shade variation
11 - Thread trimming	Trim all thread tails after wash, before final pressing/packing.	N/A	N/A	N/A	No loose tails visible on exterior

05 - BRANDING, LABELING & COLORWAYS



ID	Artwork	Specification	Placement	Size / dimension	Method	Approval
BR-01	Main label - VAEREN STUDIOS	Black woven label; logo mark + VAEREN STUDIOS wordmark; clean monochrome.	Inside back neck, centered	Proposed 55 x 18 mm	Damask/woven	ARTWORK + HANDLOOM
BR-02	Size tab	Black tab with white size letter; XS/S/M/L/XL.	Below main label, centered	Proposed 15 x 22 mm	Woven/satin	ARTWORK
BR-03	Care label	Composition, care symbols, country of origin, manufacturer/legal info as required by destination market.	Left side seam, 5-8 cm above hem	Final size by copy length	Satin print	LEGAL COPY
BR-04	Optional back-neck micro mark	Small cross/star icon shown in concept board; use only if included in approved artwork package.	Under/near main label	Max 8 mm symbol	Woven/jacquard or print	BRAND APPROVAL
BR-05	Packaging logo	VAEREN STUDIOS mark centered on frosted polybag.	Front of polybag	Approx. 100-140 mm wide	1C print	PACKAGING

Artwork handoff: factory receives vector logo/artwork separately. Do not redraw the logo from the image. Label dimensions shown here are proposed and must be signed off on the actual handloom.

06 - WASH / FINISH / COLOR STANDARD

Target effect: garment-washed, low-contrast vintage surface with controlled marbling/abrasion. The wash must not compromise rivet retention, zipper function, seam integrity or measurement stability.

Colorway	Base shade direction	Wash / finish	Hardware	Thread	Status	Notes
CW-01 WASHED BLACK	Deep charcoal-black; no blue cast	Garment washed; controlled vintage abrasion	Matte gunmetal / black	Match shell	PRIMARY	Must reproduce board's low-contrast marbled surface
CW-02 WASHED GREY	Mid-dark cool grey	Same wash recipe adjusted by shade	Matte gunmetal	Match shell	CONCEPT	Use same surface character, not identical recipe if dye differs
CW-03 WASHED BROWN	Dark cocoa / brown-black	Garment washed; low contrast	Antique gunmetal	Match shell	CONCEPT	Keep tonal and premium; avoid orange cast
CW-04 WASHED OLIVE	Dark desaturated olive	Garment washed; low contrast	Antique gunmetal	Match shell	CONCEPT	Avoid yellow/bright military olive

Wash development step	Required control	Record / approve
1. Lab dip / shade	Match primary CW-01 washed black; no blue cast; compare under consistent light.	Shade card + photo + physical swatch
2. Pilot wash	Lock detergent, enzyme/stone/abrasion level, time, temperature, liquor ratio and dryer recipe.	Factory wash recipe v1
3. Shrinkage	Measure M and one adjacent size pre/post wash. Update grade if needed.	Pre/post POM report
4. Surface	No excessive whisker/abrasion concentration; no holes unless intentionally sampled and approved.	Physical wash swatch
5. Hardware	No zipper corrosion/staining, no loose rivets, no fabric pull-through.	Hardware/wash test report

07 - PACKAGING, QC & TESTING

Check	Requirement	Tolerance / pass condition	Method	Owner	Proto	Bulk	Notes
Fabric weight	400-450 GSM target after approved finish	Within approved lab standard / commercial tolerance	Fabric GSM test; ASTM D3776/D3776M	Factory	OPEN	OPEN	Confirm whether target is before or after wash, lock this in lab report
Shade	Matches approved lab dip / wash standard	No unacceptable panel-to-panel or lot mismatch	Visual + lightbox if available	QC	OPEN	OPEN	Approval under consistent lighting
Shrinkage	Garment wash must be controlled	Target dimensional change per approved wash report; proposed $\pm 3\%$	Pre/post wash measurement	Factory/QC	OPEN	OPEN	Update once wash recipe is locked
Pilling / abrasion	Premium surface with controlled wash wear	No unacceptable early pilling or abrasion for intended use	AATCC/ASTM method selected by factory/customer	QC	OPEN	OPEN	Test method to be agreed in PO
Seam strength	All stress seams secure	No seam opening during agreed test	Appropriate seam/garment test	QC	OPEN	OPEN	Focus arm panels, shoulder and collar
Rivet retention	Rivet must not tear panel fabric	No spinning, burrs or fabric pull-through	Manual + pull/torque test defined by factory	Factory	OPEN	OPEN	Approve force threshold before production
Zipper function	Smooth open/close; no snag	100% functional; no wave or twist	Manual 10-cycle test per garment	QC	OPEN	OPEN	Reject damaged or misaligned zippers
Measurement	A-E per graded POM	Within listed tolerance	Tape measure on relaxed garment	QC	OPEN	OPEN	Measure after final wash/finish
Appearance	No stains, needle damage, oil marks, raw thread, asymmetric panels	AQL/customer standard	Final inspection	QC	OPEN	OPEN	Add destination-market AQL in PO
Labeling	Correct size, care, composition, origin	100% correct and legible	Visual / copy check	QC	OPEN	OPEN	Legal review before production
Packing	Fold and polybag without crushing hood/panels	Consistent pack method	Packing audit	Warehouse	OPEN	OPEN	Do not compress riveted panels

Standards note: Suggested reference methods in the QC sheet should be selected and locked in the purchase/order documentation; customer specifications and destination-market rules take precedence.

08 - APPROVAL GATES & RELEASE CONTROL

Gate	Deliverable	Acceptance criteria	Status	Approved by	Date	Version	Comments
01	Pattern / technical flat	Panel geometry and POMs match approved design	OPEN				
02	Proto 1	Fit, hood shape, sleeve mobility, zipper function acceptable	OPEN				
03	Wash test	Shade, hand, shrinkage and surface effect match approved standard	OPEN				
04	Hardware	YKK zipper + rivet approved for finish, strength and appearance	OPEN				
05	Size set	XS-XL gradation approved after fit review	OPEN				
06	PP sample	Final construction and branding match approved pack	OPEN				
07	Bulk release	All QC and legal requirements complete	HOLD				

RELEASE RULE

Bulk production is HOLD until Pattern, Proto, Wash, Hardware, Size Set and PP approvals are recorded. Any pattern or wash change after PP requires written revision and re-approval.

REFERENCE STANDARDS / RELIABLE SOURCES

ISO 3758:2023 - Textiles - Care labelling code using symbols. Official ISO page: <https://www.iso.org/standard/74401.html>

ASTM D5034-21(2025) - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test). Official ASTM page: <https://store.astm.org/d5034-21r25.html>

AATCC textile test methods directory, including TM61 colorfastness to laundering. Official AATCC page: <https://www.aatcc.org/testing/standards>

ASTM textile standards directory, including D3776/D3776M for fabric mass per unit area and D4966 for abrasion. Official ASTM page: <https://store.astm.org/products-services/standards-and-publications/standards/textile-standards.html>

Prepared from: the supplied VAEREN STUDIOS V.02 Armored Essential Hoodie design board. No factory pattern file, vector artwork file, approved wash recipe or material test report was supplied in the source image.